Online Sales Analysis

1. Project Overview and Objective

Overview:

This project involves cleaning, transforming, and analysing raw data using Excel and creating an interactive Power BI dashboard to derive meaningful business insights.

Object:

The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to make informed decisions.

2. Data Sources

- **Source Description and Timeline:** https://www.zoho.com/analytics/sample-data-generator.html
- Domain: sales

3. Problem Statement;

- To analyse sales performance across products to identify high performing areas and profit trends.
- To study crop production patterns across online shopping sales.
- To understand customer purchasing behaviour to enhance marketing and retention strategies.

4. Attribute (Column /Features) Details:

1. Product Details

S.No	Attribute Name	Description	Data Type

1	Transaction _ID	Unique ID for each transaction	Text
2	Product _ ID	Unique ID for each product	Text
3	Product_category	Category of the product	Categorical
4	Product_name	Name of the product	Text
5	Quantity	Number of units purchased	Numerical
6	Unit_Price	Price of one product unit	Numerical
7	Total_Amount	Total sales amount	Numerical

2. Customer Details

S.No	Attribute Name	Description	Data Type
1	Customer_ID	Unique ID for each customer	Text
2	Product_ID	ID of the purchased product	Text
3	Custmer_name	Name of the customer	Text
4	Custmer_Email	Email address of the customer	Text

3. Transaction Details

S.No	Attribute Name	Description	Data Type

1	Transaction_ID	Unique ID for each transaction	Text
2	Product_ID	ID of the purchased product	Text
3	Transaction_Date	Date and time of the transaction	Date/Time
4	Payment_Status	Status of the payment	Categorical

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard.

6. Data Pre-Processing (Excel / Power Query)

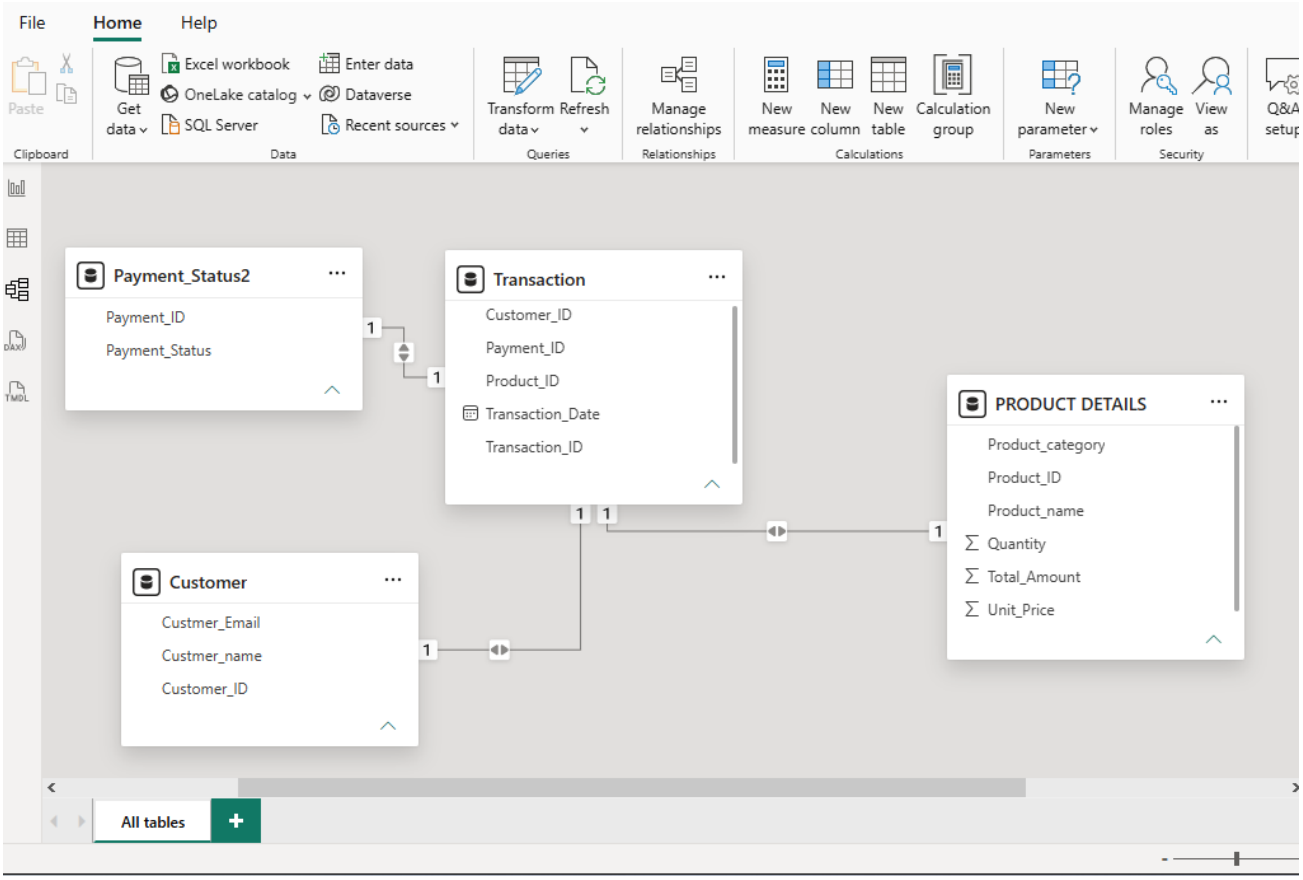
Tasks Performed:

- **Data Cleaning & Transformation:** create separate id for product , customer , make separate sheets using id.
- **Filtering & Sorting:** Organized data to focus on relevant records.

7. Data Modelling and DAX (Power BI)

- Data Model: Established relationships between tables, defined cardinality, and created lookup tables where necessary.

SCREENSHOT OF DATA MODELLING



DAX Measures:

Implemented DAX formulas for key metrics, such as total sales, AVG sales , high sales , min sales and total unit.

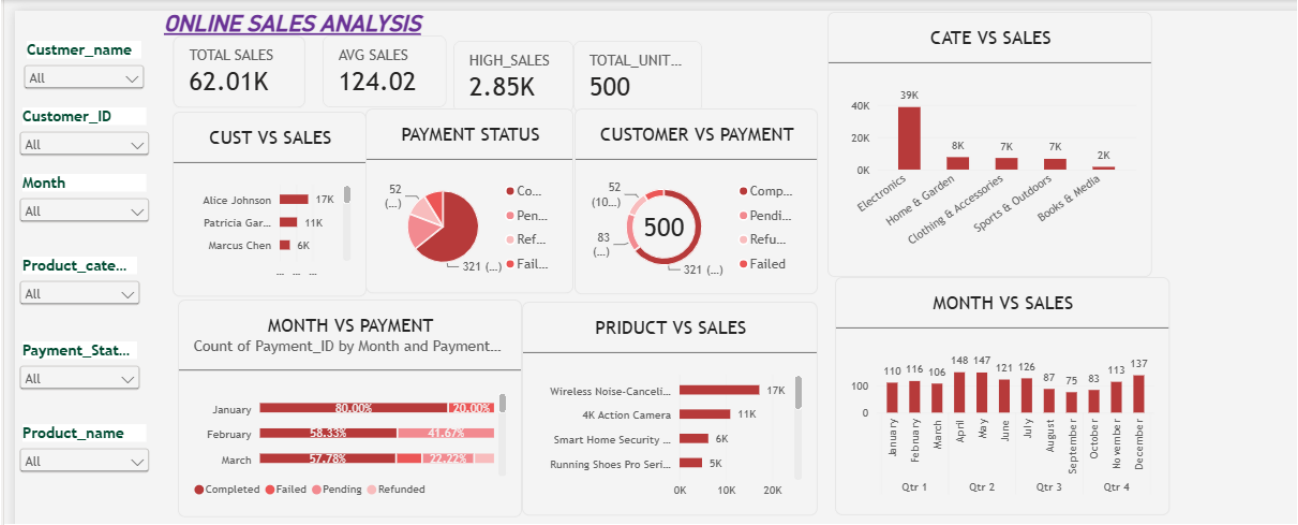
FORMULA:

- **Total sales formula:** total_ sales = SUM('PRODUCT DETAILS'[Total_ Amount])
- **Average sales :** average _sales = AVERAGE('PRODUCT DETAILS'[Total _ Amount])
- **High sales :** HIGH_SALES = CALCULATE(SUM('PRODUCT DETAILS'[Total_ Amount]),FILTER('PRODUCT DETAILS','PRODUCT DETAILS'[Total _Amount]>500))
- **Min sales:** MIN_SALES = MIN('PRODUCT DETAILS'[Total_ Amount])
- **Total unit sales:** TOTAL_UNIT_SALES = COUNT('PRODUCT DETAILS'[Quantity])

7. Analysis and Visualizations (Power BI):

- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, and tables to communicate insights.
- Make your Report interactive Layout using **Drill-down**, **Filters and slicers.**
- **Bookmarks, Clear Titles for each visuals & Labels is must.**
- **Create a consolidated Report /Dashboard.**

SCREENSHOT OF DASHBOARD /REPORT



9. Insights & Conclusions

- **Key Findings:**

- The dashboard helps identify total sales and average sales.
- Customer-wise analysis shows which customers contribute more to sales.
- Category-wise analysis helps identify popular product categories.
- Product-wise analysis helps compare product sales performance.
- Payment status analysis helps understand successful and pending payments.
- Monthly analysis helps identify sales trends over time.

- **Provide the analysis insights:**

1. Descriptive Analysis

- Analyzed total sales, average sales, minimum sales, and maximum sales.
- Identified sales performance by customers, products, and categories.

2. Diagnostic Analysis

- Compared sales across different customers and product categories.
- Analyzed payment status to understand payment-related differences.
- Examined monthly sales variations to understand changes in performance.

3. Predictive Analysis

- Used historical monthly sales trends to understand possible future sales patterns.
- Identified potential changes in sales based on previous trends.

4. Prescriptive Analysis

- Focus on products and categories with lower sales to improve performance.
- Improve customer engagement to increase sales.

10. Conclusions:

The Product Sales Analysis project helps understand overall sales performance using Power BI. It provides useful insights into customer behaviour, product categories, monthly sales trends, and payment status. Interactive dashboards and visualizations make it easier to identify sales patterns and business performance. This analysis helps businesses make better decisions, improve sales, and plan future strategies.